Unit 1: Project Planning & Management

1.1 Project Proposal

1.1.1 Overview

Moodly is an advanced, multi-platform mental health application developed using the **Flutter framework**, chosen for its high performance in creating a seamless cross-platform experience. The project is designed to offer a holistic digital environment that addresses the psychological and emotional needs of users through a modern, user-centered interface. A key aspect of the project's development was the use of **Figma** for UI/UX prototyping, which allowed the team to refine the "Safe Haven" visual theme and soothing palettes before implementation. By integrating AI-driven interaction, professional therapy tools, and lifestyle tracking, Moodly aims to reduce barriers to mental health support and encourage daily self-care habits.

1.1.2 Project Objectives

The primary objective of Moodly is to deliver a reliable, data-driven tool for mental health management. Key goals include:

- **Emotional Awareness:** Enabling users to visualize patterns through specialized mood calendars and interactive charts.
- **Holistic Wellness:** Bridging the gap between physical habits and mental states through integrated water and nutrition tracking.
- **On-Demand Support:** Providing instant emotional assistance via an AI-powered chatbot and a dedicated meditation resource hub.
- **Professional Accessibility:** Streamlining the process of finding and booking licensed therapists through a secure, transparent system.
- **Social Connectivity:** Building a safe community platform for shared experiences and peer support.

1.1.3 Project Scope

The scope of Moodly covers extensive modules designed for high performance on both Android and iOS. The technical infrastructure utilizes **Supabase** as a robust backend solution for managing real-time data and user authentication. The scope includes personalized onboarding questionnaires, a therapeutic media suite (Meditation, Podcasts, and ASMR), and lifestyle management tools (Water and Food tracking). Additionally, it encompasses a professional hub for therapist reviews and session bookings, secure payment integration, and a real-time AI Chatbot. The development process is strictly managed via **GitHub** to ensure efficient version control and collaborative coding among the five team members.

1.2 Project Plan

The projectfollowsthe **Agile methodology**, which was selected to ensure flexibility and iterative improvement throughout the development lifecycle. This approach allowed the team to work on different modules in parallel, facilitating a more efficient development process. Coordination among team members was maintained through constant communication via **WhatsApp and Zoom**, ensuring all technical and design requirements were met.

1.2.1 Development & Collaboration Tools

To ensure high-quality execution and effective team collaboration, the following tools were utilized:

- **Design Tool (Figma):** Used for comprehensive UI/UX design, wireframing, and creating high-fidelity interactive prototypes.
- **Development Framework (Flutter & Dart):** Selected for building a high-performance, cross-platform mobile application for Android and iOS from a single codebase.
- **Backend-as-a-Service (Supabase):** Integrated as the primary backend for real-time database management, secure user authentication, and file storage.
- **Version Control (GitHub):** Employed for collaborative coding, managing software versions, and tracking changes across the development team.
- **Communication Tools (WhatsApp & Zoom):** Used for daily coordination, weekly sync-up meetings, and rapid decision-making.

1.2.2 Project Timeline

The following timeline outlines the key phases and milestones of the project, highlighting the strategic decision to prioritize UI/UX design and frontend development before backend integration:

Phase	Duration	Description
Planning & Requirements	Jan 6 – Jan 25	Defining project architecture, features, and system flow.
UI/UX Design & Prototyping	Jan 26 – Feb 20	Developing 29+ high-fidelity screens and flows in **Figma**.
Frontend Development (Flutter)	Feb 21 – Mar 31	Implementing visual components and navigation in **Flutter**.
Backend & Logic Integration	Apr 1 – Apr 30	Connecting **Supabase**, authentication, and payment
Testing & Optimization	May 1 – May 15	Conducting user acceptance testing and performance tuning.
Final Submission	May 25	Completion of final documentation and system deployment.

1.3 Task Assignment & Roles

Responsibilities were distributed to ensure parallel development across all key modules:

- **Karim Motaz (Team Leader):** Oversees the project roadmap. Technically responsible for the **Home Screen** and its dynamic sections, alongside the Meditation Module.
- **Ahmed Hamdy:** Focuses on the **Authentication Logic** and the **Payment Infrastructure**, managing both backend security in **Supabase** and UI integration.
- **Amira Yasser:** Specialized in the **AI Chatbot** ecosystem, developing the interactive interface and the underlying emotional support logic.

- **Karin Maged:** Responsible for the **User Management** modules, including the **Profile** system and essential legal frameworks **(Terms and Policies)**.
- **Yassin Yahia:** Manages the **Social Connectivity** through the **Community Screen** and ensures the performance of the **Media Player** for multimedia content.

1.4 System Architecture & Data Management

The application relies on a robust relational database schema hosted on **Supabase (PostgreSQL)**. This schema is designed to support the app's diverse features while ensuring data integrity and security. Key components include:

- **User & Identity Management:** Managed through user_data and questionnaire_answers, ensuring a personalized journey from the first login.
- **Therapeutic Ecosystem:** A relational structure between therapists, availability, and time slots, which facilitates the bookings and payments workflow.
- **Communication Infrastructure:** Real-time messaging is supported via chat rooms and messages tables, allowing secure user-therapist interaction.
- **Wellness Tracking:** The moods table tracks emotional history, while podcasts and sound therapy tracks manage the distribution of relaxation content.
- **Subscription Logic:** A dedicated subscriptions system distinguishes between standard and premium access to features like ASMR and advanced analytics.

1.5 Risk Assessment & Mitigation Plan

- **Technical Integration:** Potential challenges with AI and Payment APIs are mitigated by early prototyping and rigorous testing phases.
- **Data Privacy:** Sensitive user data and therapist records are protected using **Supabase** secure authentication and data encryption protocols.
- **System Consistency:** To maintain UI/UX integrity, the team adheres strictly to the predefined **Figma** design system throughout the Flutter implementation.

1.6 Key Performance Indicators (KPIs)

Moodly's success is measured by **System Reliability** (backend uptime), **User Engagement** (daily tracking frequency), and **Response Accuracy** (AI Chatbot efficiency). The application aims for a seamless performance with a load time of under 3 seconds to ensure a frictionless therapeutic experience.

Unit 2: Literature Review

2.1 Feedback & Evaluation

- The Moodly project demonstrates a well-structured and practical mobile application in the field of mental health support. The system integrates multiple useful features such as mood tracking, therapist booking, real-time chat, and multimedia content including podcasts and sound therapy.

- From a technical perspective, the database design is clear and normalized, covering essential modules such as users, therapists, bookings, payments, and messaging. The inclusion of tables like moods, bookings, chat rooms, and subscriptions reflects a solid understanding of real-world application requirements.

- The application also shows good consideration for user experience by providing features like questionnaire-based personalization and therapist ratings. Additionally, the separation between user roles (users, therapists, admin) is well-handled.

- Overall, the project is functional, relevant, and demonstrates a good level of software engineering practices suitable for a graduation-level system.

2.2 Suggested Improvements

While the project is well implemented, several improvements can enhance its quality without adding major complexity:

- Improve error handling in booking and payment processes to ensure better reliability.
- Add simple input validation on user data (e.g., phone, email format) to enhance data integrity.
- Optimize database queries for frequently accessed data such as therapist listings and chat messages.
- Enhance the UI consistency across different screens for a more unified user experience.
- Add a basic notification system (e.g., booking confirmation or reminders).
- Improve the rating system by displaying average ratings for therapists.
- Add simple loading indicators to improve perceived performance.

These improvements are incremental and can be implemented without significant architectural changes.

2.3 Final Grading Criteria

The project will be evaluated based on the following criteria:

- Documentation: 20%
- Implementation (Functionality & Features): 30%
- Testing & Reliability: 20%
- User Interface & User Experience: 15%
- Presentation & Demonstration: 15%

Total: 100%

Unit 3: Requirements Gathering

3.1 Stakeholder Analysis

3.1.1 Overview

The Moodly ecosystem focuses on three primary groups that drive the platform's core interactions:

Stakeholder	Primary Role	Interest	Influence
End Users	Individuals seeking mental health support & tracking.	Very High	Medium
Therapists	Licensed professionals providing clinical sessions.	High	High
Admin	System managers overseeing content, users, and quality.	High	Very High

3.1.2 Key Stakeholder Profiles

- **End Users (Patients):** Need secure mood tracking, therapist access (Chat/Live), and AI support. Their main goal is improving well-being with full data privacy.
- **Therapists / Counselors:** Need tools to manage profiles, session schedules, and patient reviews. Their goal is efficient practice management.
- **Administrators:** Need to verify therapists, monitor payments, and manage wellness content (Podcasts/Meditation).

3.1.3 External Service Dependencies

To ensure high performance, the system integrates the following:

- **Supabase:** For Authentication, Database, and Storage.
- **Stripe & Paymob:** For global and local payment processing.
- **ZegoCloud:** For real-time video therapy sessions.

3.2 User Stories & Use Cases

This section categorizes the core interactions within the Moodly system, mapping user needs to functional priorities.

Module	Functional Summary (User Goals)	Priority
Auth & Onboarding	Secure registration/login (Email/Google), password recovery, and completing a personalization questionnaire. Returning users skip onboarding.	Must Have
Mood Tracking	Logging daily emotional states and visualizing trends through interactive weekly/monthly history charts.	Must Have
Therapist & Booking	Browsing professional profiles, viewing reviews, and booking Chat/Live sessions via an integrated availability calendar.	Must Have
Therapy Sessions	Conducting real-time private video calls and text-based messaging. Facilitating therapist tools for schedule management.	Must Have
Reviews & Ratings	Submitting, editing, or deleting therapist feedback to ensure transparency and community trust.	Should Have
Wellness Content	Accessing a library of meditations, podcasts, and sound therapy (ASMR), with full library access restricted to Premium Users.	Must Have
AI Chatbot	Engaging with an AI-driven companion for 24/7 immediate emotional support, coping strategies, and mental health tips.	Must Have
Community	A social layer for sharing personal experiences, reading peer posts, and providing support via likes and comments.	Should Have
Health Tracking	Monitoring daily water intake (Hydration stats) and receiving personalized meal recommendations tailored to the user's mood.	Should Have
Payments	Secure transaction processing for individual sessions and premium subscriptions via international (Stripe) and local (Paymob) gateways.	Must Have
Settings & Profile	Managing personal data, app ratings, secure logout, and UI customization (Dark/Light themes).	Should Have

3.3 Functional Requirements

This section defines the core capabilities of the **Moodly** system, categorized by functional domains to ensure a seamless mental health support experience.

Domain	Core System Capabilities (Must/Should Have)
User & Access	Secure **Email/Google OAuth** authentication, password recovery, and session persistence. Includes an **onboarding questionnaire** for new users with a skip logic for returning users.
Mood Tracking	Logging emotional states with timestamps and visualizing data through **dynamic history charts** (Daily/Weekly/Monthly) using emoji-based indicators.
Therapy Booking	A searchable directory with detailed therapist profiles (bio, pricing, ratings). Includes a **dynamic slot generation** system to prevent double-booking.
Consultations	**Real-time communication** via ZegoCloud for Live video/audio sessions and Supabase real-time subscriptions for secure text-based chat.
Content Library	Centralized media player for **Meditation, ASMR, and Podcasts**. Features include category filtering and **Subscription-based locking** for premium audio content.
AI Support	An **AI-powered Chatbot** capable of multi-turn, empathetic conversations to provide immediate coping strategies and mental health advice.
Reviews & Feedback	Integrated system for users to rate and review therapists (with an anonymous option) and provide overall app feedback to the administrators.
Community Feed	A moderated social space where users can publish posts, share experiences, and engage via **likes and comments**.
Wellness Tracking	Tools for **hydration monitoring** (water intake logs) and an AI-driven engine for **meal recommendations** tailored to wellness goals.
Payment & Subs	Automated billing via **Stripe (International)** and **Paymob (Local)**. Includes automated validation of subscription status (Active/Expired) on app launch.

3.4 Non-functional Requirements

This section defines the quality attributes of the **Moodly** system, ensuring high performance, robust security, and an optimal user experience across all platforms.

Category	Quality Attribute & Technical Target	Metric / Target
Performance	App cold start, API responses, and media (audio) playback must be fast and responsive. Real-time chat should have minimal latency.	< 3s Load / < 1s Chat Latency
Security	Authentication via Supabase Auth (bcrypt). Data in transit encrypted via TLS 1.2+. Use of Row-Level Security (RLS) for data isolation.	100% RLS & TLS Enforced
Privacy & Data	Zero storage of credit card data (delegated to Stripe/Paymob). Secure storage of tokens using Flutter Secure Storage. GDPR-compliant data deletion flows.	PCI & GDPR Compliance
Usability	Consistent UI across iOS/Android. Support for Light/Dark modes. Navigation must reach any primary feature in ≤ 3 taps.	WCAG 2.1 Contrast Standards
Reliability	High system availability and uptime. Graceful handling of network failures with informative offline states. Atomic booking/payment operations.	≥ 99.5% Uptime
Scalability	Support for horizontal scaling via PostgreSQL connection pooling. Media delivered via CDN to support high traffic without bottlenecks.	10,000+ Concurrent Users
Maintainability	Codebase follows Clean Architecture. State management via BLoC/Cubit. JSON serialization and Dependency Injection (GetIt) are enforced.	BLoC & Clean Architecture

Unit 4: System Analysis & Design

4.1 Problem Statement & Objectives

4.1.1 Problem Statement

Many users struggle to understand and track their emotional patterns and maintain healthy daily habits. This lack of awareness negatively affects their mental well-being.

Solution:

Moodly is a comprehensive system that helps users track their mood, improve lifestyle habits, access mental health resources, and engage with a supportive community.

4.1.2 Objectives

- Enable users to track their mood daily.
- Provide personalized recommendations (meditation, audio, food).
- Facilitate therapist booking and communication.
- Improve lifestyle habits (water intake, nutrition).

4.1.3 Functional & Non-Functional Requirements

Functional Requirements

- Users can register and log in.
- Users can track their mood.
- Users can view personalized recommendations.
- Users can chat and join communities.
- Users can book therapy sessions.
- Users can make payments.
- Users can use chatbot assistance.

Non-Functional Requirements

- Performance: Fast response time (< 2 seconds).
- Security: Data encryption and secure authentication.
- Scalability: Modular and extensible architecture.
- Usability: Simple and user-friendly UI.
- Reliability: Stable system with minimal crashes.

4.1.4 Software Architecture

The system follows Clean Architecture, which ensures scalability and maintainability.

Layers:

Presentation Layer:

- UI (Screens).
- State Management (Cubit / Bloc).

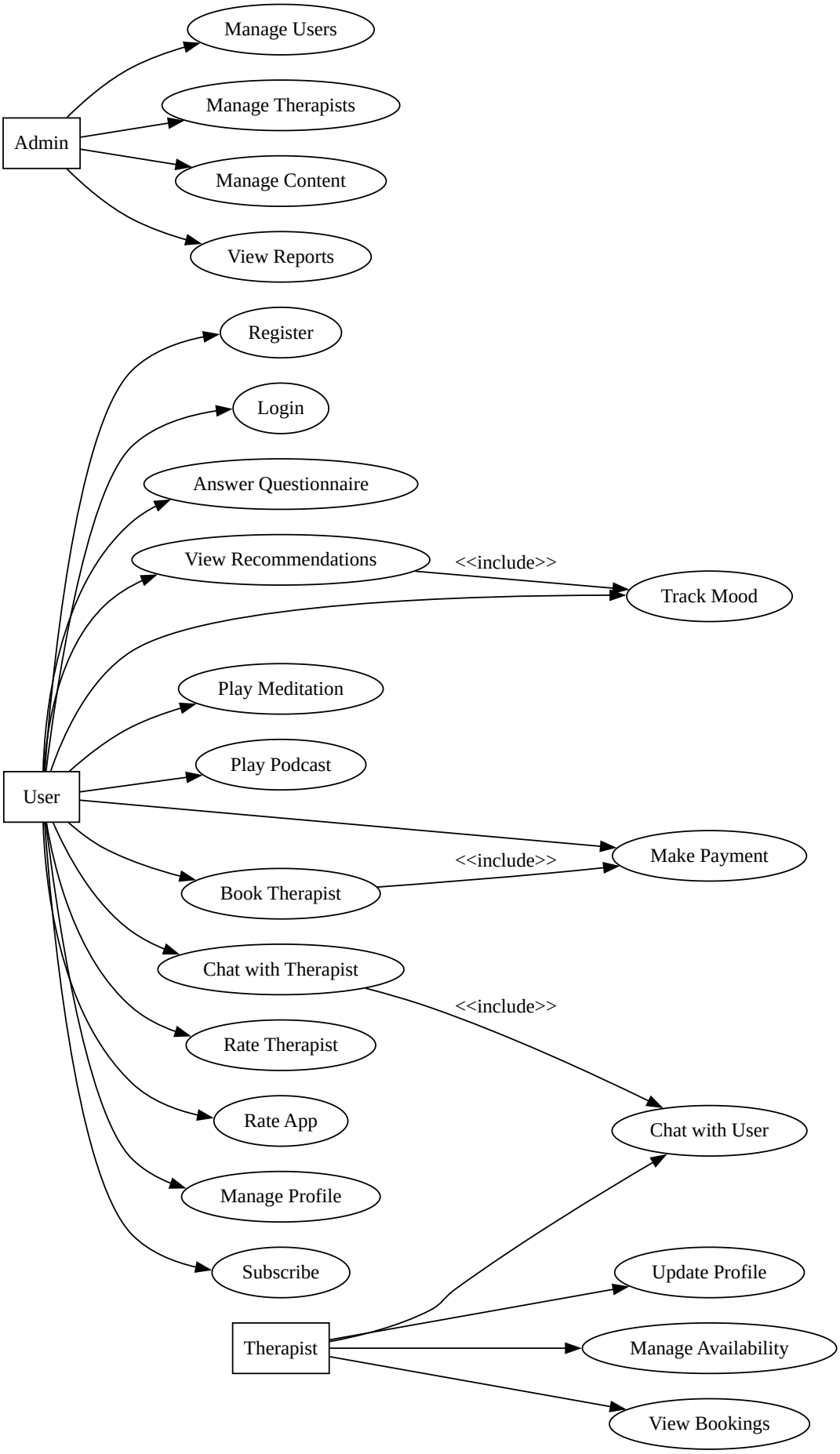
Domain Layer:

- Entities.
- Use Cases.

Data Layer:

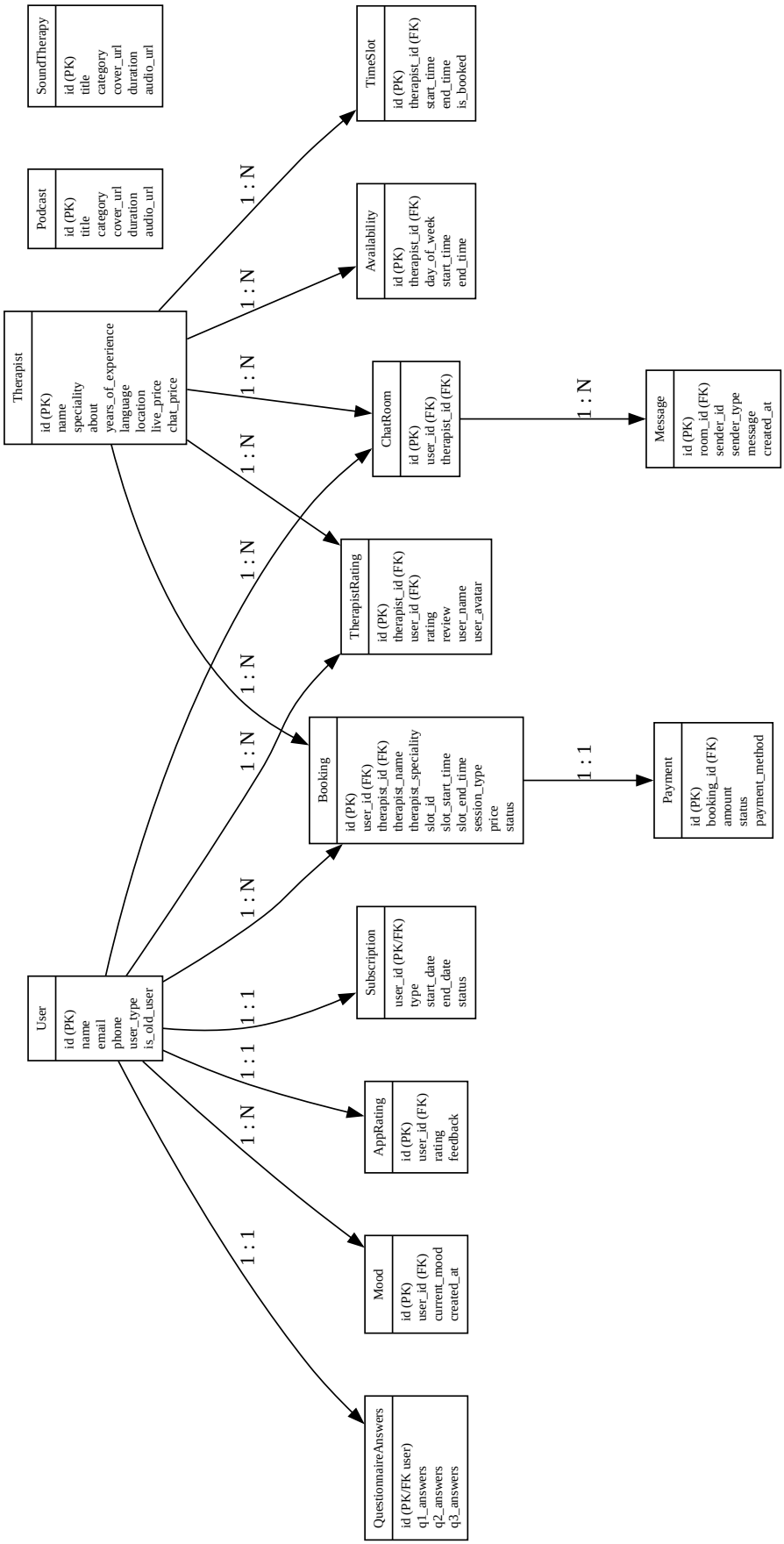
- Repositories
- API Services (e.g., Supabase)

4.1.5 Use Case Diagram & Descriptions



4.2 Database Design & Data Modeling

4.2.1 ER Diagram (Entity-Relationship Diagram)



4.2.2 Logical & Physical Schema

1.Logical Schema (Conceptual Design)

The logical schema represents the high-level structure of the system data, independent of any specific database implementation. It defines the main entities, their attributes, and relationships.

User Entity
- user(id, name, email, phone, user_type, is_old_user)

Therapist Entity
- therapist(id, name, speciality, about, years_of_experience, language, location, live_price, chat_price, discount)

Mood Tracking
- mood(id, user_id, current_mood, created_at)

Questionnaire
- questionnaire_answers(user_id, q1_answers, q2_answers, q3_answers)

Booking System
- booking(id, user_id, therapist_id, slot_start_time, slot_end_time, session_type, price, status)

Payment System
- payment(id, booking_id, amount, payment_method, status, created_at)

Chat System
- chat_room(id, user_id, therapist_id)
- message(id, room_id, sender_id, sender_type, message, created_at)

Rating System
- therapist_rating(id, user_id, therapist_id, rating, review)
- app_rating(id, user_id, rating, feedback)

Content System
- podcast(id, title, category, audio_url, duration)
- sound_therapy(id, title, category, audio_url, duration)

Subscription
- subscription(user_id, type, start_date, end_date, status)

2. Physical Schema (Database Implementation)

The physical schema represents the actual database structure implemented in PostgreSQL, including data types, constraints, primary keys, and foreign keys.

- **Example: User Table**

```
CREATE TABLE user_data (
  id uuid PRIMARY KEY,
  name text,
  email text,
  phone text,
  user_type text DEFAULT 'user',
  is_old_user boolean DEFAULT false
);
```

- **Example: Booking Table**

```
CREATE TABLE bookings (
  id uuid PRIMARY KEY,
  user_id uuid NOT NULL,
  therapist_id uuid NOT NULL,
  slot_start_time timestamp NOT NULL,
  slot_end_time timestamp NOT NULL,
  session_type text NOT NULL,
  price numeric,
  status text DEFAULT 'pending'
);
```

Primary Keys (PK)

- All tables use id as the primary key except:
- subscription → uses user_id

Foreign Keys (FK)

- booking.user_id → user_data.id
- booking.therapist_id → therapists.id
- mood.user_id → user_data.id
- payment.booking_id → bookings.id
- message.room_id → chat_rooms.id

3. Normalization Considerations

The database design follows normalization principles to ensure data consistency and reduce redundancy.

First Normal Form (1NF)

- All attributes contain atomic (indivisible) values
- No repeating groups or arrays inside tables
- Each field contains a single value

Second Normal Form (2NF)

- All non-key attributes are fully dependent on the primary key
- No partial dependency exists in composite structures

Third Normal Form (3NF)

- No transitive dependencies between non-key attributes
- Each attribute depends only on the primary key

Controlled Denormalization (Design Decision)

Some controlled redundancy is introduced for performance optimization:

- Booking table stores therapist snapshot (name, speciality)
- Chat messages are stored in a flat structure for real-time performance

Justification

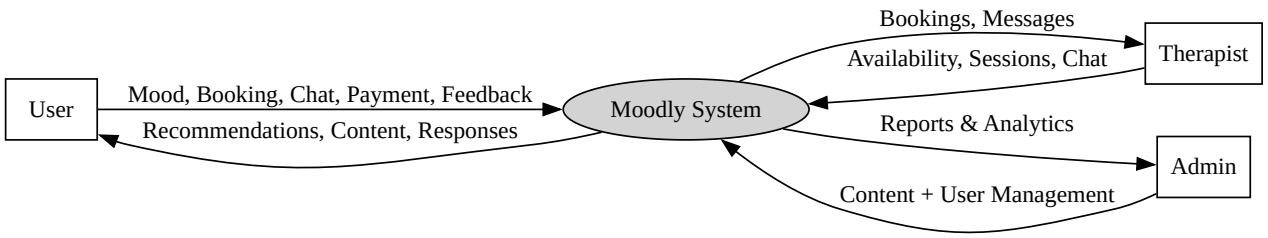
This trade-off was applied to:

- Improve query performance in mobile environments
- Reduce join complexity
- Support real-time features (chat & booking)

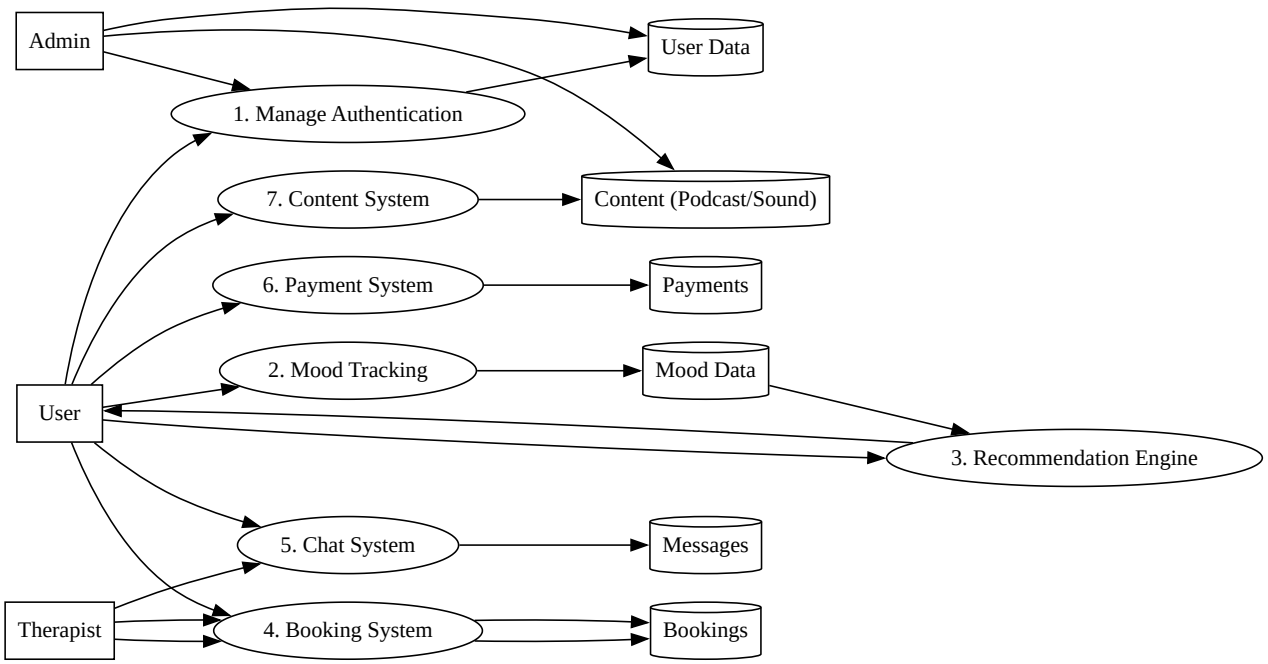
4.3 Data Flow & System Behavior

4.3.1 DFD (Data Flow Diagram)

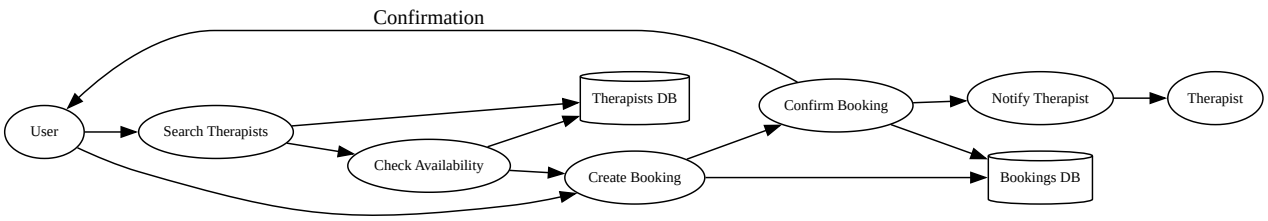
DFD Level 0 (Context Diagram)



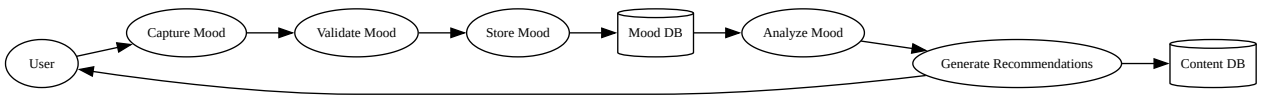
DFD Level 1 (Detailed System)



DFD Level 2 – Booking System

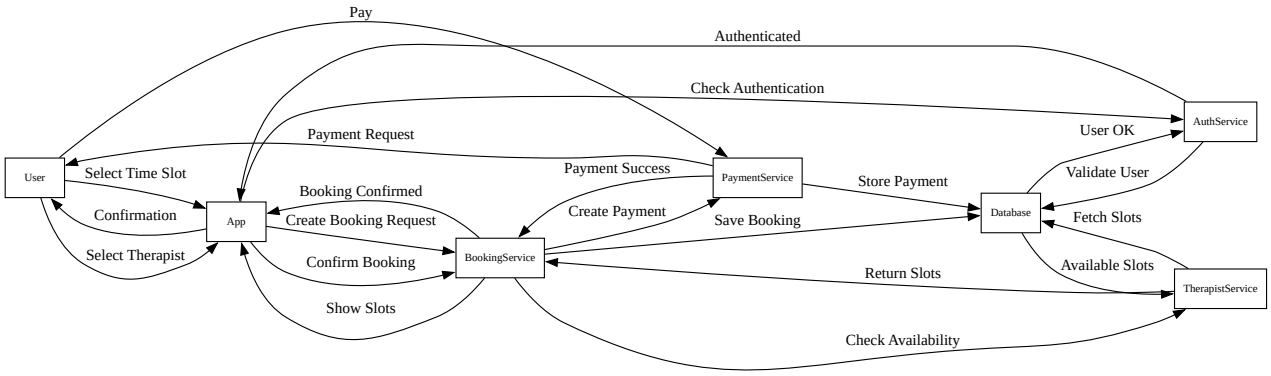


DFD Level 2 – Mood System

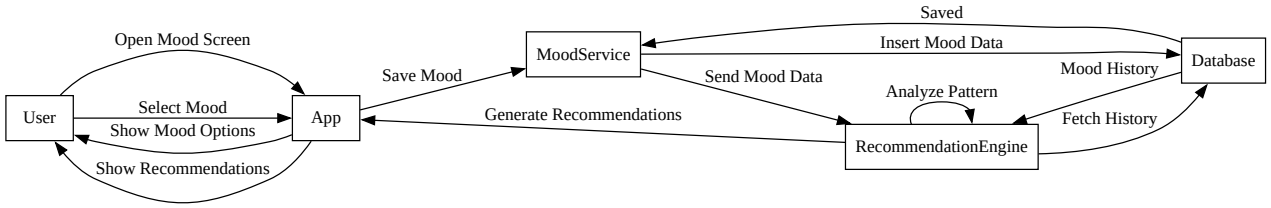


4.3.2 Sequence Diagrams

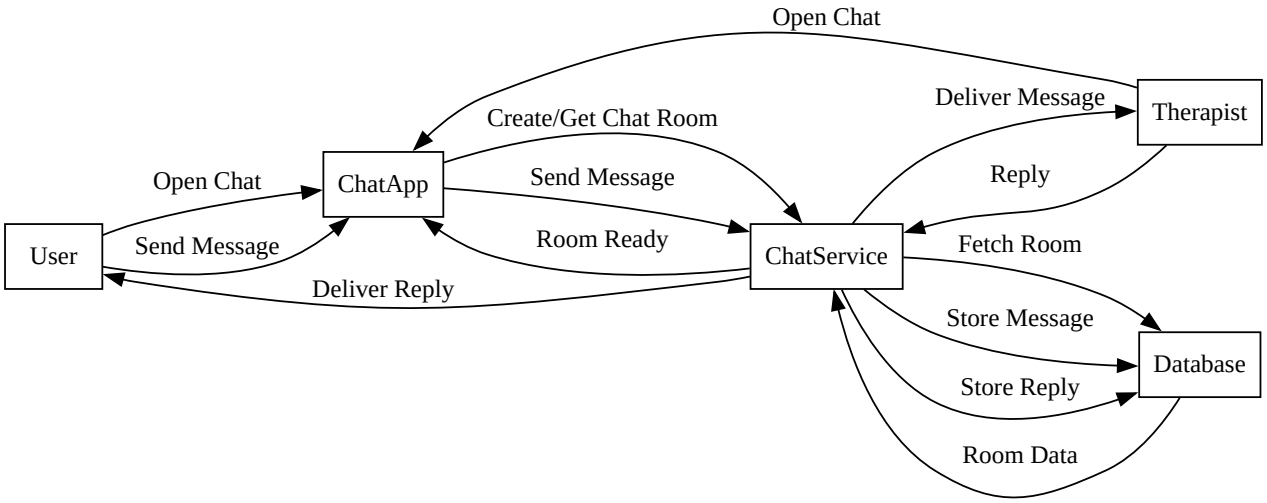
Book Therapist Flow



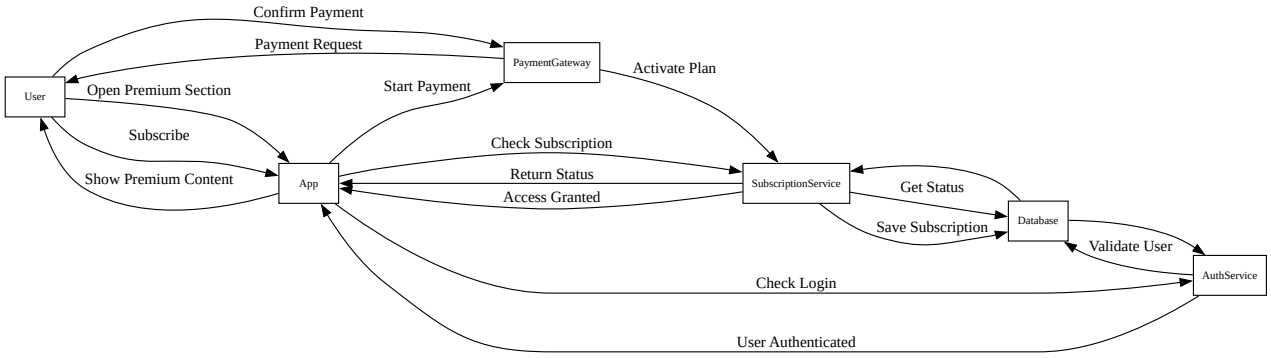
Mood Tracking + Recommendations



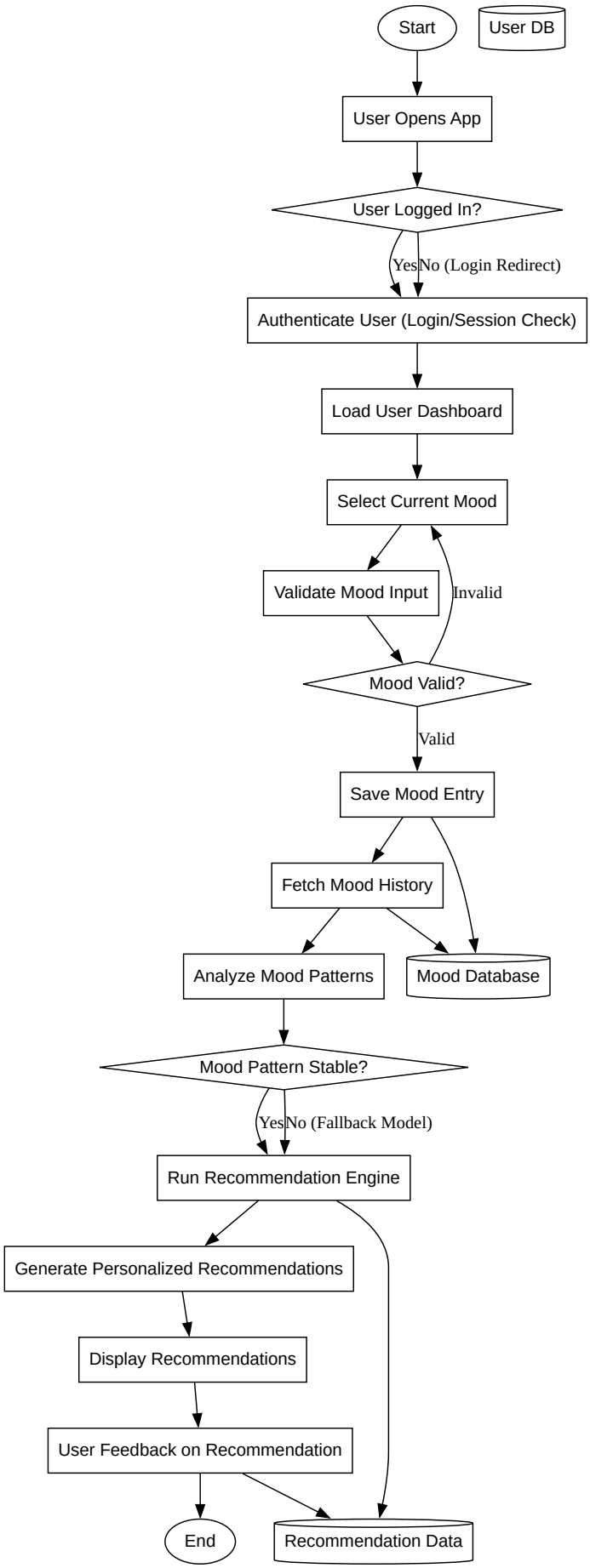
Chat System (Real-time)



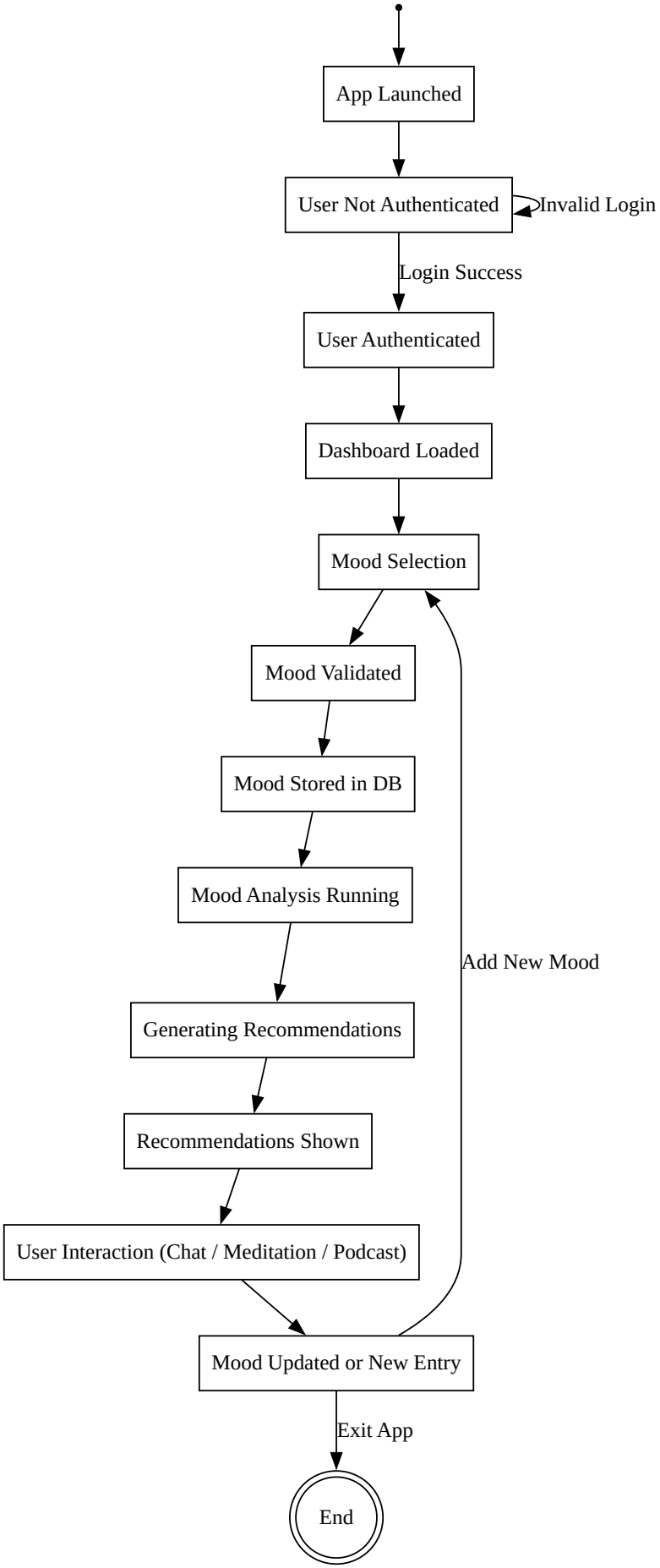
Subscription & Premium Access



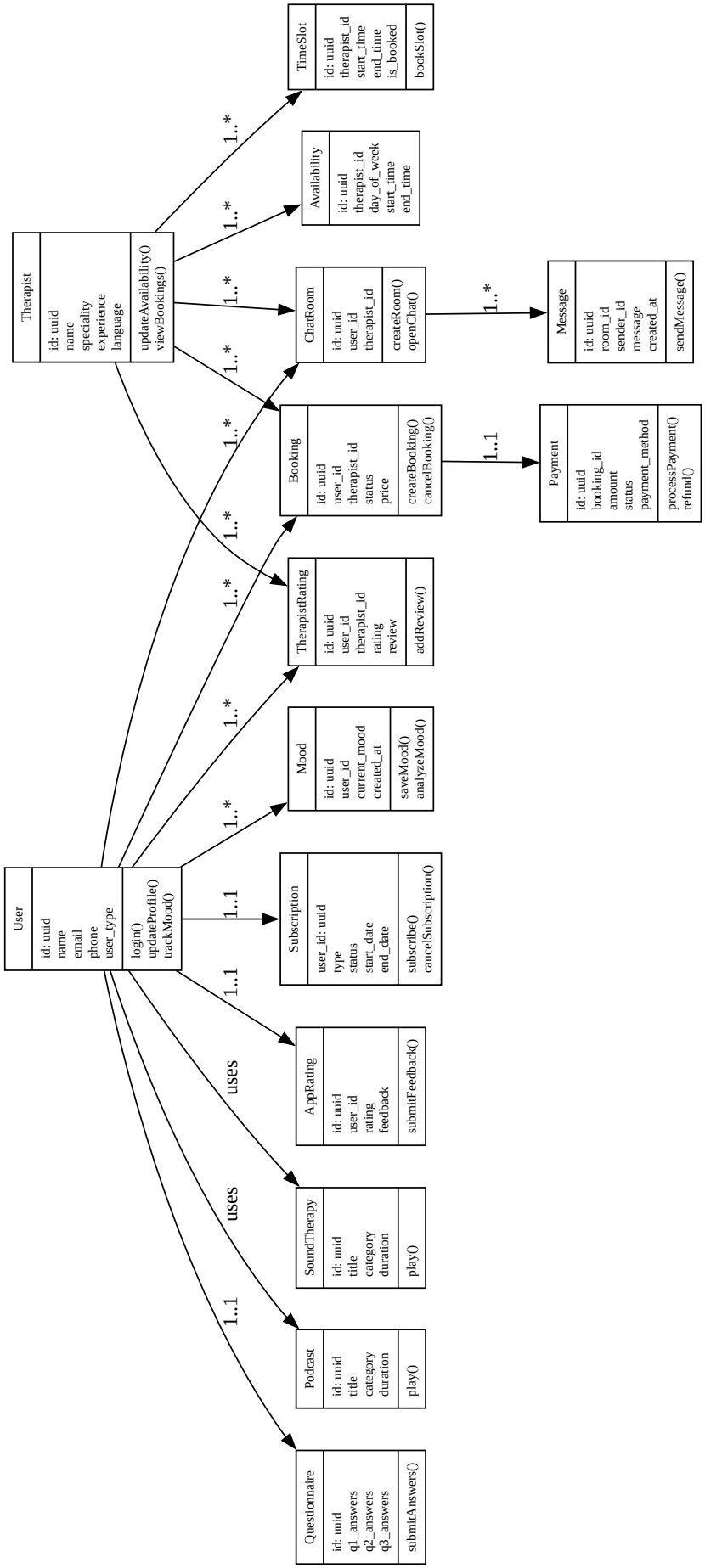
4.3.3 Activity Diagram



4.3.3 State Diagram



4.3.3 Class Diagram



4.4 UI/UX Design & Prototyping

4.4.1 Wireframes & Mockups

The UI/UX design of the Moodly application was created to ensure a simple, calming, and user-friendly experience suitable for mental health and emotional well-being users.

The system includes the following main screens:

- Splash & Onboarding Screens
- Authentication Screens (Login / Register / Forgot Password)
- Home Screen (Mood Dashboard)
- Mood Tracking Screen
- Recommendations Screen (Meditation, Podcasts, Sound Therapy)
- Booking Screen (Therapists Listing & Scheduling)
- Chat Screen (User–Therapist Communication)
- Subscription & Payment Screen
- Profile & Settings Screen

Wireframes Overview

Wireframes were designed in a low-fidelity format to focus on layout and user flow before implementing visual styling.

Key Wireframe Goals:

- Simplify navigation between features
- Reduce cognitive load for users
- Prioritize mood tracking and recommendations
- Ensure fast access to therapy and support features

Mockups (High-Fidelity Design)

High-fidelity mockups were developed to represent the final visual appearance of the application, including:

- Icons and illustrations for emotional states
- Clean card-based layouts for recommendations
- Therapist profile cards with booking buttons
- Chat interface with minimal distractions
- Audio player UI for podcasts and sound therapy

4.4.2 UI/UX Design Guidelines

The design of Moodly follows modern mental-health-centered UX principles, focusing on emotional comfort, clarity, and accessibility.

Color Scheme

The color palette was carefully selected to create a calm and relaxing environment:

- Primary Color: Soft Blue (#6C8EFF)
- Secondary Color: Lavender (#A78BFA)
- Background: Light Gray / White (#F7F8FA)
- Success: Soft Green (#4ADE80)
- Warning: Soft Orange (#FB923C)
- Error: Soft Red (#F87171)

Typography

A clean and modern font style was used:

- Primary Font: Poppins / Roboto
- Headings: Semi-Bold (600–700 weight)
- Body Text: Regular (400 weight)
- UI Labels: Medium (500 weight)

Accessibility Considerations

The system was designed with accessibility in mind:

- High contrast between text and background
- Large touch targets for mobile usability
- Simple navigation structure
- Clear iconography for emotions and actions
- Minimal text complexity for better understanding
- Support for scalable font sizes

UX Design Principles

The following UX principles were applied:

Simplicity
- Avoiding unnecessary complexity in user flows.

Emotional Design
- Using colors and visuals that support mental well-being.

Consistency
- Uniform layout structure across all screens.

User-Centered Design
- Prioritizing user emotional state and quick access to support features.

Minimal Cognitive Load
- Reducing distractions and focusing on core actions like mood tracking and therapy access.

Prototyping Approach

The prototyping phase followed an iterative design process:

- Low-fidelity wireframes (layout structure)
- High-fidelity mockups (final UI design)
- Interactive prototype (navigation flow simulation)
- User flow validation (ensuring smooth experience)

Final Outcome

The UI/UX design of Moodly ensures:

- A calm and supportive digital environment
- Easy access to mental health features
- Smooth navigation between modules
- A visually appealing and emotionally intelligent interface

Click here to view the UI/UX design prototype

4.5 System Deployment & Integration

4.5.1 Technology Stack

The Moodly system was developed using a modern full-stack architecture to ensure scalability, performance, and maintainability.

Frontend (Mobile Application)

- Framework: Flutter
- Language: Dart
- State Management: Bloc/Cubit
- UI Design: Material Design 3

Backend

- Backend Platform: Supabase
- Authentication: Supabase Auth / JWT
- APIs: RESTful APIs

Database

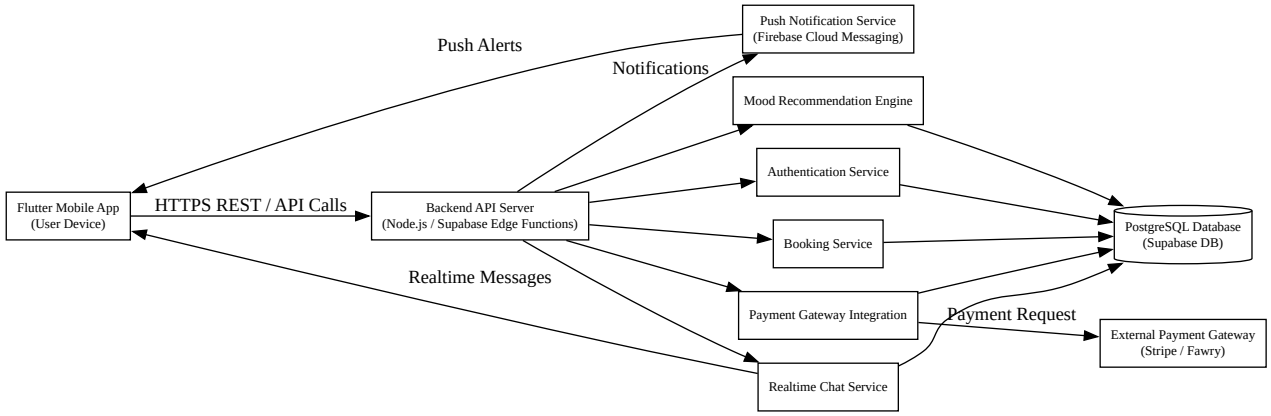
- Database System: PostgreSQL
- Hosted on: Supabase Cloud
- Features:
  - Relational schema
  - Row Level Security (RLS)
  - Real-time subscriptions

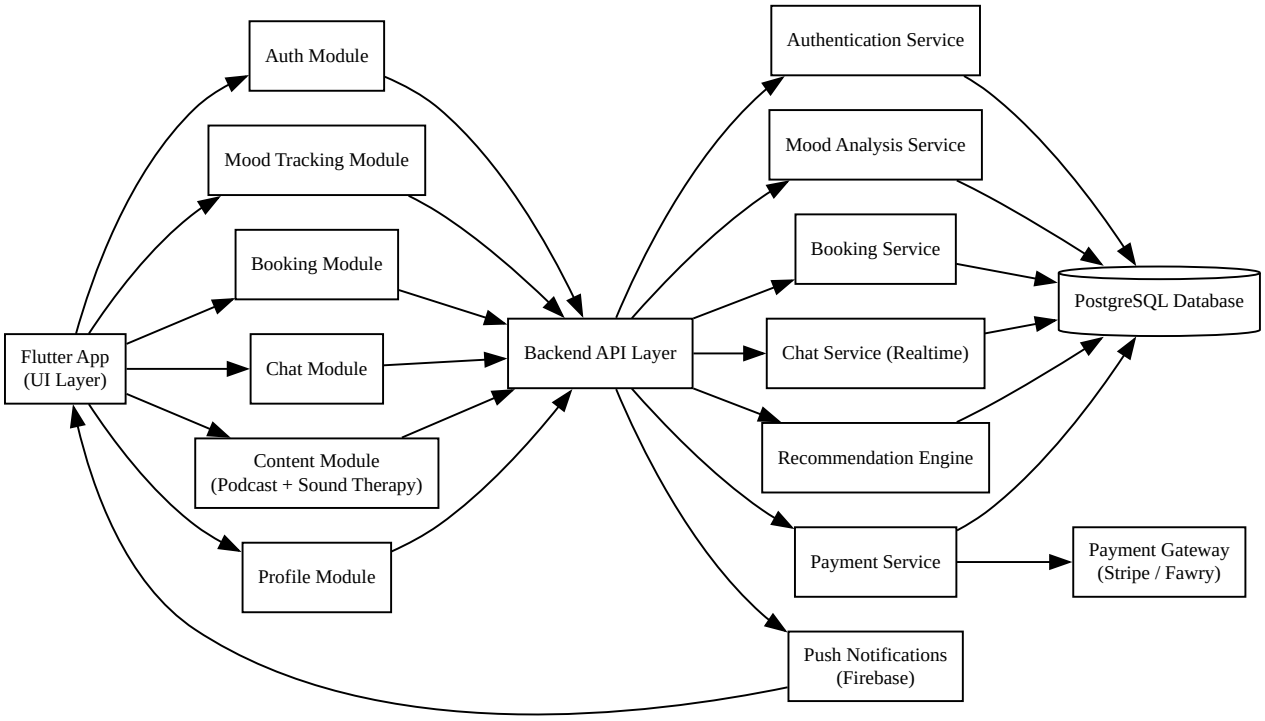
External Services

- Payment Gateway: Stripe / Paypal
- Notifications: Firebase Cloud Messaging (FCM)
- Media Storage: Cloud Storage (Supabase Storage)

4.5.2 Deployment Diagram

The deployment architecture illustrates how the system components are distributed across different hardware and cloud environments.



4.5.3 Component Diagram

The system is designed using a modular component-based architecture, where each module is responsible for a specific functionality.